

The Morse Hamiltonian: Exact Classical Action and Complete Quantum Bound Spectrum

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Abstract

We solve the classical and quantum one-dimensional Morse Hamiltonian under one convention lock. Every bounded classical orbit has an elementary action and period, while canonical quantization has a finite, explicitly exhausted generalized-Laguerre point spectrum. The proof derives both formulas and fixes the strict square-integrability cutoff.

1 Frozen Hamiltonian and result

Let

$$V(x) = D(e^{-2ax} - 2e^{-ax}), \quad H_{\text{cl}}(x, p) = \frac{p^2}{2m} + V(x), \quad (1)$$

where $m, D, a > 0$. The potential has minimum $-D$ at zero, tends to zero from below as $x \rightarrow +\infty$, and tends to $+\infty$ as $x \rightarrow -\infty$. For a closed regular energy curve we use the normalized action

$$J(E) = \frac{1}{2\pi} \oint p \, dx. \quad (2)$$

For $\hbar > 0$, let H_q be the self-adjoint operator associated with the lower-bounded closed form of

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \quad \text{on } L^2(\mathbb{R}), \quad \lambda = \frac{\sqrt{2mD}}{a\hbar}.$$

Theorem 1 (Classical action and quantum closure). *For every $-D < E < 0$, the classical energy curve is periodic and*

$$J(E) = \frac{\sqrt{2mD}}{a} \left(1 - \sqrt{-E/D}\right), \quad T(E) = \frac{2\pi}{a\sqrt{-2E/m}}, \quad J'(E) = \frac{T(E)}{2\pi}. \quad (3)$$

The essential spectrum of H_q is $[0, \infty)$. Its point spectrum is exactly the finite set

$$E_n = -\frac{a^2 \hbar^2}{2m} \left(\lambda - n - \frac{1}{2}\right)^2, \quad n \in \mathbb{N}_0, \quad n < \lambda - \frac{1}{2}, \quad (4)$$

and every eigenvalue is simple. With $z = 2\lambda e^{-ax}$, a corresponding eigenfunction is, up to normalization,

$$\psi_n(x) = z^{\lambda-n-1/2} e^{-z/2} L_n^{(2\lambda-2n-1)}(z). \quad (5)$$

There are $\max(0, \lceil \lambda - 1/2 \rceil)$ bound states. Equality in the cutoff produces a zero-energy constant tail, not an L^2 eigenfunction; there are no nonnegative eigenvalues.

The phrase *Morse classical action owner* is the round-zero certificate: it fixes the model, the 2π normalization, and the strict spectral statement before any finite computation.

2 Classical proof

Put $y = e^{-ax}$ and $\delta = \sqrt{1 + E/D} \in (0, 1)$. The two turning values of y are $1 - \delta$ and $1 + \delta$; their order in x is reversed. A full oscillation has period

$$\begin{aligned} T(E) &= \sqrt{2m} \int_{x_-}^{x_+} \frac{dx}{\sqrt{E - V(x)}} \\ &= \frac{\sqrt{2m}}{a\sqrt{D}} \int_0^\pi \frac{d\theta}{1 + \delta \cos \theta} = \frac{\pi\sqrt{2m}}{a\sqrt{-E}}. \end{aligned}$$

The last integral follows directly from the tangent-half-angle substitution. Differentiating (2) under the integral gives $J'(E) = T(E)/(2\pi)$; endpoint terms vanish because $p = 0$ at both turning points. Since the curve contracts to the equilibrium at $E = -D$, we have $J(-D) = 0$. Integrating the period from $-D$ to E proves all of (3). This derivation also detects an erroneous use of the unnormalized closed integral as the action.

3 Quantum reduction and exhaustion

Write a prospective negative eigenvalue as $E = -(a^2\hbar^2/2m)s^2$ with $s > 0$. The change $z = 2\lambda e^{-ax}$ turns the Schrödinger equation into

$$z^2\psi_{zz} + z\psi_z + (-z^2/4 + \lambda z - s^2)\psi = 0. \quad (6)$$

Substitution $\psi = z^s e^{-z/2} F$ gives

$$zF'' + (2s + 1 - z)F' + (\lambda - s - \tfrac{1}{2})F = 0. \quad (7)$$

At $z \rightarrow \infty$, square integrability selects the recessive Whittaker solution. Its small- z nonintegrable coefficient is proportional to $1/\Gamma(s - \lambda + 1/2)$, and hence vanishes precisely when

$$\lambda - s - \tfrac{1}{2} = n \in \mathbb{N}_0.$$

Then (7) terminates as $L_n^{(2s)}(z)$. Since $dx = -(az)^{-1}dz$, the remaining endpoint test is $\int_0 z^{2s-1} dz < \infty$, exactly when $s > 0$. Thus the inequality in (4) is strict and the displayed list exhausts all negative eigenvalues. Sturm oscillation supplies simplicity and node count n . The special-function conventions agree with the official references [2, 3].

References

- [1] P. M. Morse, *Diatomic Molecules According to the Wave Mechanics. II. Vibrational Levels*, Phys. Rev. 34 (1929), 57, [doi:10.1103/PhysRev.34.57](https://doi.org/10.1103/PhysRev.34.57).
- [2] NIST Digital Library of Mathematical Functions, *Laguerre polynomials*, <https://dlmf.nist.gov/18.5>.
- [3] NIST Digital Library of Mathematical Functions, *Whittaker functions*, <https://dlmf.nist.gov/13.14>.